

MAINCRAFT INTERNSHIP PROGRAM

Python Programming Internship Task-1

Core Python challenges

① Python program to get sum of two numbers.

Code :-

take two numbers as input from user

a = int(input("Enter number 1:"))

b = int(input("Enter number 2:"))

Sum = a + b

Print ("the sum of numbers is :")

Output

Enter Number 1: 4

Enter Number 2: 6

the sum of numbers is : 10.

② Python program to check if a number is odd/even

Odd or Even Checker

```
Code :- num = int(input("Enter a number:"))  
if num % 2 == 0 :  
    print(num, "is Even")  
else :  
    print(num, "is Odd")
```

Output

→ Enter a number : 12
12 is Even

=====

Enter a number : 7
7 is Odd

1 2 3 4 5 6 7 8 9 10 11 12

→ : 1 2 3 4 5 6 7 8 9 10 11 12

2 : 1 2 3 4 5 6 7 8 9 10 11 12

0 : 1 2 3 4 5 6 7 8 9 10 11 12

② Python program to check if a number is odd/even

Odd or Even Checker

```
Code :- num = int(input("Enter a number :"))  
if num % 2 == 0 :  
    print(num, "is Even")  
else :  
    print(num, "is Odd")
```

Output

→ Enter a number : 12
12 is Even

= - - - - -

Enter a number : 7
7 is Odd

③ Python program to calculate factorial

Factorial Calculation

Code :- `num = int(input("Enter a number :"))
factorial = 1`

`if num < 0:`

`print("Factorial does not exist for
 negative numbers.")`

`else :`

`for i in range(1, num+1):`

`factorial = factorial * i`

`print("Factorial of", num, "is", factorial)`

Output

Enter a number : 5

factorial of 5 is 120

④ Program to generate fibonacci series

Fibonacci Sequence

Code : — `n = int(input("Enter the number of terms:"))`

`a = 0`

`b = 1`

`for i in range(n):`

`print(a, end=" ")`

`c = a + b`

`a = b`

`b = c`

Output

Enter the number of terms : 10

0 1 1 2 3 5 8 13 21 34

⑧

String Reverse

Reverse user-input string

```
Code :- text = input("Enter a string :")
```

```
reverse = text[::-1]
```

```
print("Reversed string :", reverse)
```

Output

Enter a string : Hello

Reversed string : olleH

⑥

Palindrome Check

Is the word same forward & backward.

Code :- word = input("Enter a word :")

if word == word[::-1]:

print("Palindrome")

else:

print("Not a Palindrome")

Output

Enter a word : madam

Palindrome

Enter a word : Hello

Not a palindrome

Enter a number : 12321

A Armstrong Number

⑦

Leap Year Check.

Code :-

```
year = int(input("Enter a year :"))  
if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):  
    ~ Print (year, "is a leap year")  
else :  
    Print (year, "is Not a leap year")
```

Output

Enter a year : 2024

2024 is a leap year

=====

Enter a year : 2023

2023 is Not a leap year.

Armstrong Number

Example 153

Code :- `num = int(input("Enter a number : "))`

`S = 0`

`t = num`

`while t > 0:`

`d = t % 10`

`S = S + d**3`

`t = t // 10`

`if S == num:`

`Print("Armstrong Number")`

`else:`

`Print("Not an Armstrong Number")`

Output

Enter a number : 153

Armstrong Number